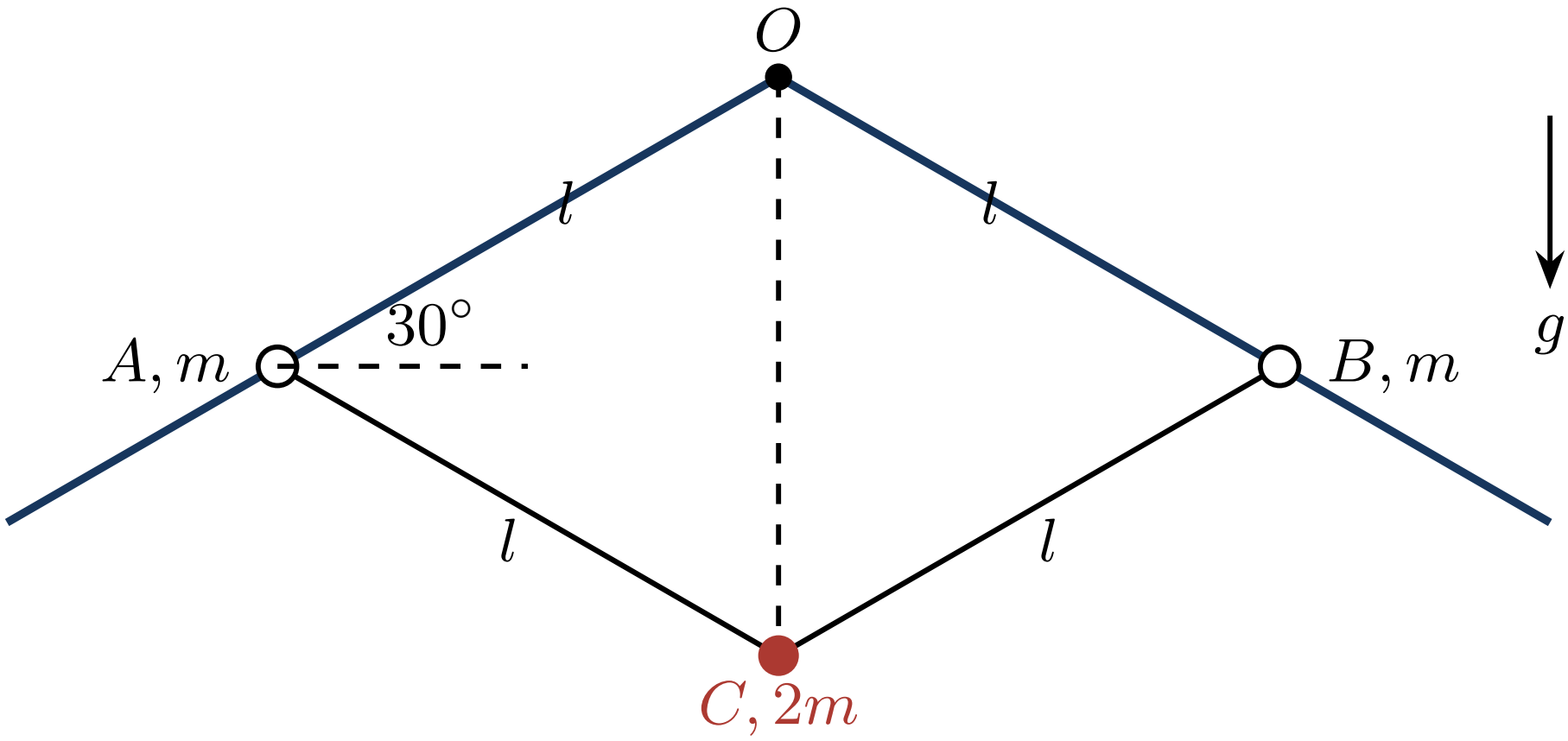
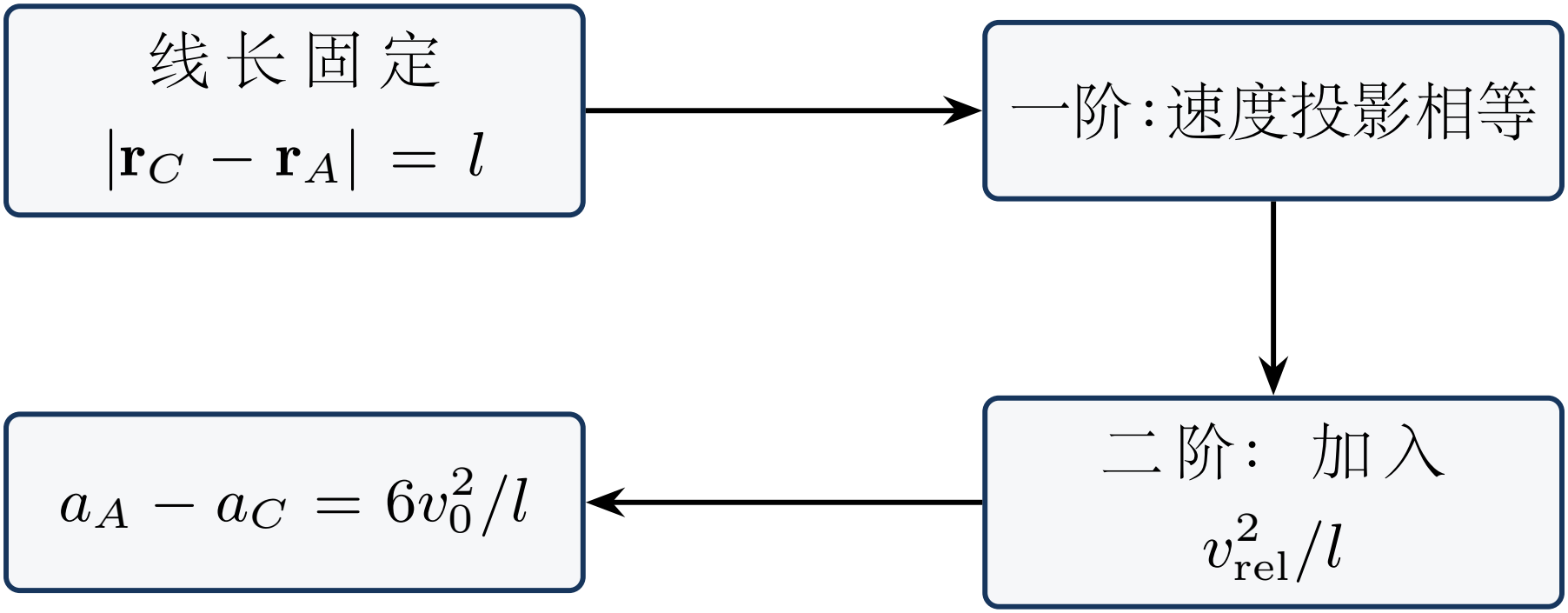


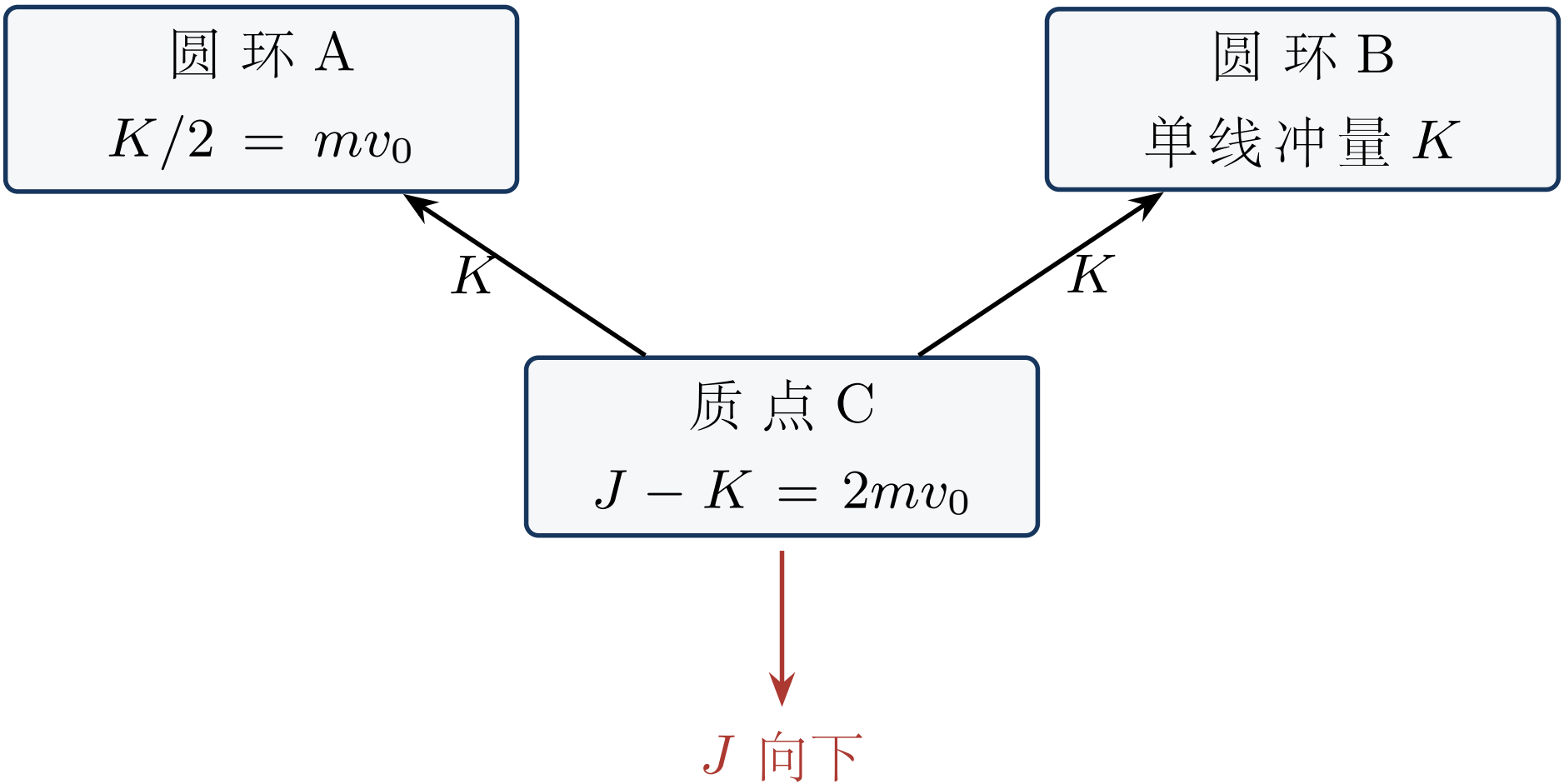
固定杆关于竖直线对称；两线绷紧，取 C 为下交点。

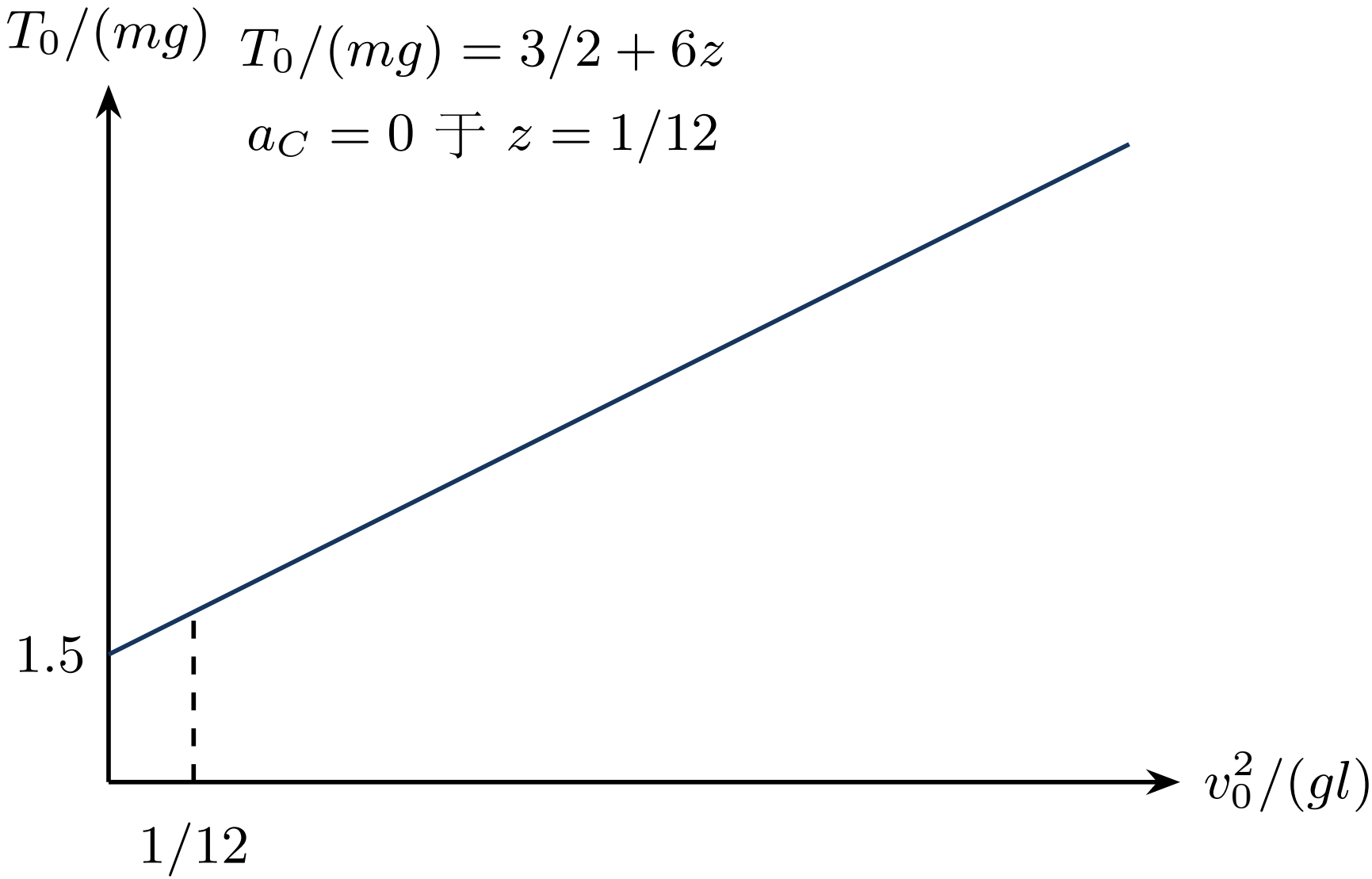


指定初始位置 $y'(l) = 1$、$y''(l) = 6/l$；静止释放时速度平方项为零。

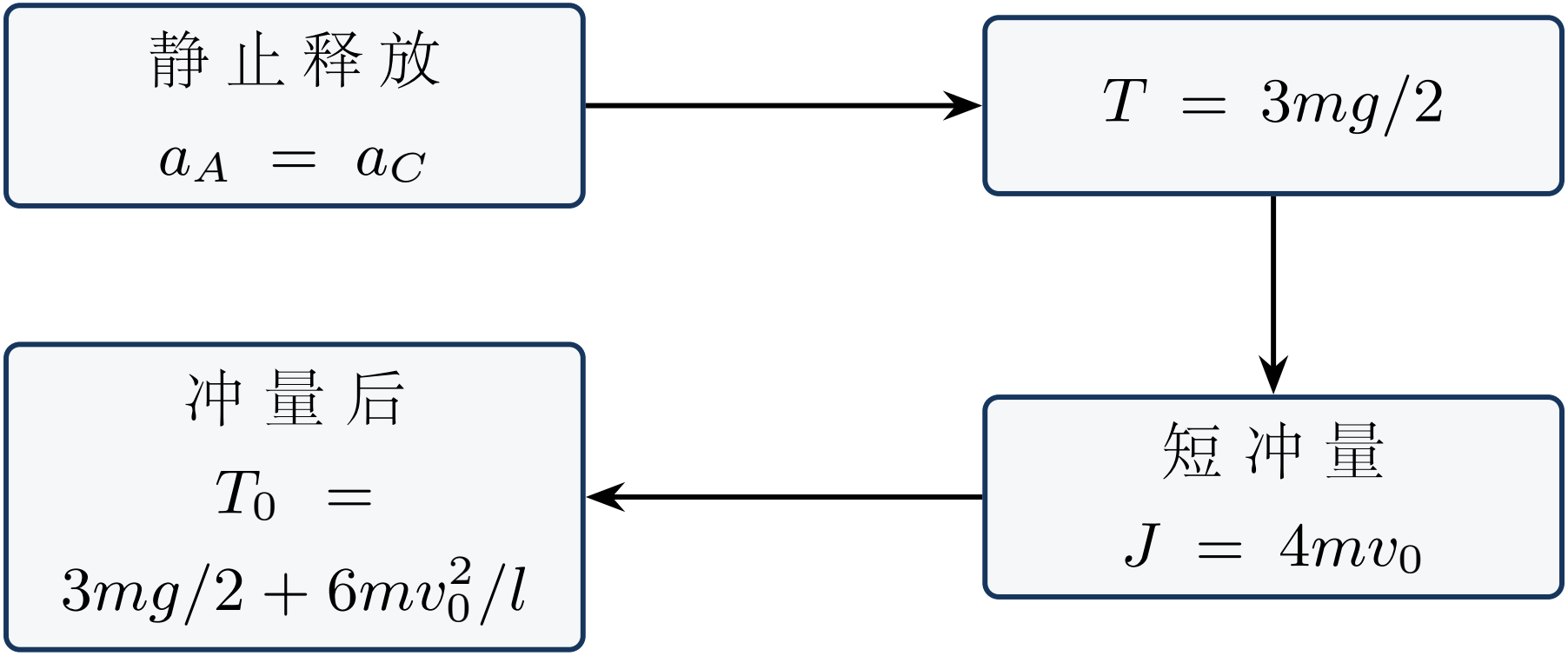
圆环沿杆速度 v_0；由 $K/2 = mv_0$

得 $K = 2mv_0$，外冲量 $J = 4mv_0$。





横轴 $z = v_0^2/(gl)$，纵轴从零起。速度向下时，加速度仍可向上。



广义等效质量 $4m$；冲量后总动能 $2mv_0^2$。